



UNIVERSITY OF MORATUWA
Faculty of Information Technology

IN2601 – COMPUTER GRAPHICS

Lab Sheet 01 : Configuring OpenGL (FreeGLUT) in Code::Blocks

Step 1: Install the Required Software

1.1 Install Notepad++

Install **Notepad++** successfully on your computer.

1.2 Run Notepad++ as Administrator

To edit system files, set Notepad++ to always run as administrator.

- Right-click **Notepad++**
- Select **Properties**
- Open the **Compatibility** tab
- Check **Run this program as an administrator**
- Click **Apply**
- Click **OK**

1.3 Install Code::Blocks

Install **Code::Blocks** successfully.

1.4 Check Compiler Detection

When Code::Blocks opens for the first time, a window named **Compilers Auto Detection** will appear.

- Make sure **GNU GCC Compiler** is shown as **Detected**
- Click **OK**

Step 2: Extract the FreeGLUT Files

2.1 Extract the ZIP File

Extract the downloaded **FreeGLUT ZIP folder**.

2.2 Identify the Main Folders

After extraction, you will see the following folders:

- **include**
- **lib**
- **bin**

Step 3: Copy Header Files

3.1 Open the Header File Folder

Go to:

freeglut → include → GL

3.2 Locate the Header Files

Inside the **GL** folder, you will find these header files:

- `freeglut.h`
- `freeglut_ext.h`
- `freeglut_std.h`
- `glut.h`

3.3 Copy the Header Files

Copy all four header files.

3.4 Paste into Code::Blocks Include Directory

Paste them into the following location:

`C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\include\GL`

Step 4: Copy Library Files

4.1 Open the Library Folder

Go to:

freeglut → lib → x64

4.2 Locate the Library Files

You will find these files:

- `libfreeglut.a`
- `libfreeglut_static.a`

4.3 Copy the Library Files

Copy both files.

4.4 Paste into Code::Blocks Library Directory

Paste them into:

`C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32\lib`

Step 5: Copy the DLL File

5.1 Open the DLL Folder

Go to:

freeglut → bin → x64

5.2 Locate the DLL File

You will find:

- `freeglut.dll`

5.3 Copy the DLL File

Copy the file.

5.4 Paste into Windows Directory

Paste it into:

`C:\Windows`

Step 6: Modify the `glut.cbp` Template File

6.1 Open the Template Folder

Go to:

`C:\Program Files\CodeBlocks\share\CodeBlocks\templates`

6.2 Find the File

Locate the file named:

- `glut.cbp`

6.3 Open with Notepad++

Open `glut.cbp` using **Notepad++**.

6.4 Replace `glut32` with `freeglut`

- Press **Ctrl + F**
- Select the **Replace** tab
- In **Find what**, type: `glut32`
- In **Replace with**, type: `freeglut`
- Click **Replace All**

6.5 Save the File

Save the file and close Notepad++.

Step 7: Modify the `wizard.script` File

7.1 Open the Wizard Folder

Go to:

```
C:\Program  
Files\CodeBlocks\share\CodeBlocks\templates\wizard\glut
```

7.2 Find the File

Locate the file named:

- `wizard.script`

7.3 Open with Notepad++

Open `wizard.script` using **Notepad++**.

7.4 Replace `glut32` with `freeglut`

- Press **Ctrl + F**
- Open the **Replace** tab
- In **Find what**, type: `glut32`
- In **Replace with**, type: `freeglut`
- Click **Replace All**

7.5 Save the File

Save the file and close Notepad++.

Step 8: Create a New GLUT Project in Code::Blocks

8.1 Open Code::Blocks

Launch **Code::Blocks**.

8.2 Create a New Project

- Go to **File** → **New** → **Project**
- Select **GLUT Project**
- Click **Next**

8.3 Enter Project Details

Provide:

- Project title
- Folder location
- Other required details

Then click **Next**.

8.4 Select GLUT Location

A window will appear asking for **Please select GLUT's location**.

- Click the **Browse** button
- A small window called **Global Variable Editor** may appear
- Click **Close**
- A message will appear: **Are you sure you want to save an invalid global variable?**
- Click **No**

8.5 Select the Correct Folder

Now select this location:

`C:\Program Files\CodeBlocks\MinGW\x86_64-w64-mingw32`

8.6 Finish Project Creation

- Click **Next**
- Click **Finish**

Step 9: Build and Run the Project

9.1 Open the Source File

In the left-side **Projects** panel:

- Expand **Sources**
- Click on `main.cpp`

9.2 Build and Run

Click **Build and Run**.

Step 10: Verify the Configuration

If the **GLUT Shapes Demo** window appears, the **FreeGLUT configuration in Code::Blocks** has been completed successfully.